2017 IEEE 56th Annual Conference on Decision and Control (CDC)
December 12-15, 2017, Melbourne, Australia

Detection of covert attacks on cyber-physical systems by extending the system dynamics with an auxiliary system

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Abstract— **Securing cyber-physical systems is vital for our modern society since they are widely used in critical infrastructure like power grid control and water distribution. One of the most sophisticated attacks on these systems is the covert attack, where an attacker changes the system inputs and disguises his influence on the system outputs by changing them accordingly. In this paper an approach to detect such an attack by extending the original system with a switched auxiliary system is proposed. Furthermore, a detection system using a switched Luenberger observer is presented. The effectiveness of the proposed method is illustrated by a simulation example.**

I. Introduction

The term cyber-physical system (CPS) was first used by Edward A. Lee [1] and describes a system that interacts with both the virtual networking world and the physical world. The focus of this paper is on the security of industrial control systems (ICS), which are a class of cyber-physical systems used to measure and control procedures in the manufacturing industry and in critical infrastructures, such as energy and water.

ICS have long been thought to be immune to cyber attacks, due to their proprietary hard- and software and lack of outside connections. One of the first papers to show that SCADA systems are not immune to attacks is [2]. One examples for recent attacks on ICS is the Stuxnet worm [3]. These considerations lead to two different fields of cyber security research for ICS, namely the detection of attacks and the design of resilient systems. This paper focuses on the detection of cyber attacks.

A good overview of different attacks can be found in [4]. [5] introduces false data injection attacks and shows that an attacker can bypass bad measurements detection techniques with knowledge of the system configuration. In [6] the influence of three different kinds of stealthy deception attacks on the control error is explored. As one of the most sophisticated attacks, covert attacks are first introduced in [7]. In the covert attack an attacker adds his attack signal to the control input of the plant. Simultaneously he simulates the influence of his attack signal on the plant outputs. These simulated outputs are then used to compensate the attacker's influence on the outputs.

In this paper covert attacks will be considered. One approach to detect covert attacks is for the defender to gain more knowledge of the system dynamics than the attacker. In [8] an extension of the plant with time-variant external states is introduced to detect covert attacks by altering the system dynamics in a way that the attacker can not correctly identify the system anymore. [9] constructs a time-varying modulation matrix to change the system inputs in order to detect covert attacks and zero dynamics attacks.

The idea of the proposed detection scheme is to obtain this superior knowledge by extending the original plant with an auxiliary system. With these dynamics unknown, the attacker can not completely compensate his influence on the system outputs. The system extension or auxiliary system is designed as a switched system which changes its dynamics at random time instants to prevent an attacker from obtaining a perfect system model of the auxiliary system. This switching signal is generated with a pseudo random number generator with the same seed in the auxiliary system and the detection system. By using an auxiliary system no additional delay is introduced into the system, which would be present if encryption is used to secure the actuation and sensing signals.

In comparison with [8], in this paper the system dynamics are extended by a discrete-time switched system, which is a more structured design method. By taking the original system into account in the design of the auxiliary system, it can be disguised as part of the original plant. Furthermore only a limited amount of matrices needs to be designed for the switched system. With a finite number of subsystems and a switching at random time instances among them a non repeating dynamic can be achieved. In [8] direct access of the plant states is required for the coupling of the systems. In the proposed approach the plant states are substituted by the plant outputs for the coupling of the auxiliary system with the original system. This is a more realistic scenario since the plant states are normally not directly accessible.

The main novelties in this paper are as follows. A structured approach to design an auxiliary system to detect a covert attack. To reduce the design efforts, the auxiliary system has a limited number of subsystems. Solutions to real world implementation problems like the synchronized switching and the lack of access to the system states for the coupling with the original system are given and due to the location of the auxiliary system it is protected from cyber attacks.

II. Problem formulation and design objectives

The first objective is the design of the linear discrete-time auxiliary system. They should be designed in such a way that they are similar to the dynamics of the original system. In case the system dynamics are analyzed by the attacker, the auxiliary system's dynamics will then not stand out in

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terms of time constants and gain. For instance a system that controls the temperature of a big tank with time constants in the range of minutes coupled with an auxiliary system that reacts in the range of milliseconds will be well visible. The states and inputs of the original system should be coupled to the original system in case only the signals of the original system are attacked, which then would also lead to a change in the auxiliary system.

The second objective is the design of the detection system. For which a switched Luenberger observer is used to handle the switched dynamics of the auxiliary system. For this detection a model of the new overall system dynamics and the switching signal of the auxiliary system are needed. The residuals of the observer are then monitored for deviations from zero, which indicates an attack.

III. System model

The cyber-physical system under consideration is structured as shown in Figure 1. The plant Σ_{Sys} is assumed to be a linear discrete time-invariant system described by

$$x_{Sys}(k+1) = A_{Sys}x_{Sys}(k) + B_{Sys}u_{Sys}(k) \quad (1)$$
$$y_{Sys}(k) = C_{Sys}x_{Sys}(k) \quad (2)$$

where $x_{Sys} \in \mathbb{R}^{n_{Sys}}$, $u_{Sys} \in \mathbb{R}^{m_{Sys}}$ and $y_{Sys} \in \mathbb{R}^{p_{Sys}}$. The matrices A_{Sys}, B_{Sys} and C_{Sys} have appropriate dimensions. The system's $Controller$, the $Detection\ system$ and the I/O are all connected through the control network.

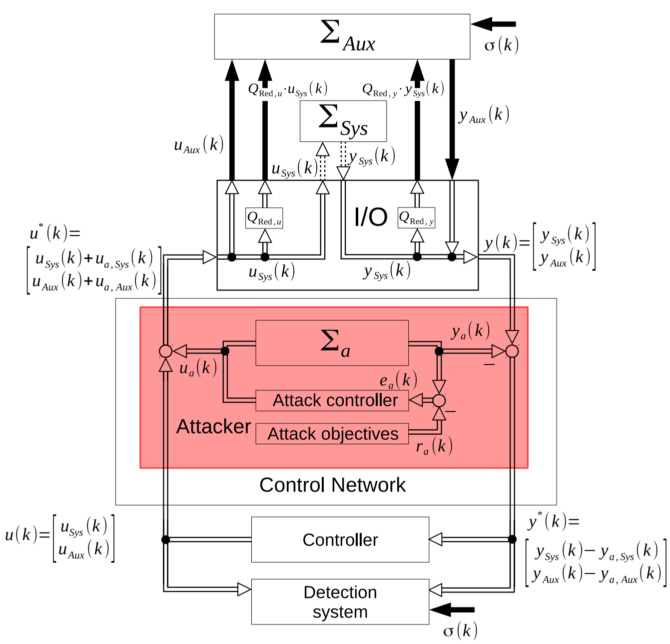

Fig. 1. Extended system with the attacker

The solid black arrows in Figure 1 represent analogue connections, the dotted lines fieldbus or analogue connections and the double solid connections outside the I/O represent the connections in the control network. The signals depicted in Figure 1 are explained throughout the rest of this paper. Here the attack is depicted as an attack on the network, but it can also be implemented as an attack on the I/O or the PLC.

IV. Design of the auxiliary system

In the first part of this section the matrices for the auxiliary system are introduced and some properties determined. The exact derivation of the matrices is done in Section IV-B. The auxiliary system is coupled via a limited number of analogue connections to the I/O of the original system and through that to the control network. These analogue connections have the advantage of being impossible to manipulate through the network and therefore the auxiliary system itself can not be attacked from afar.

A. Structure of the auxiliary system

The auxiliary system is designed as a linear discrete-time switched system. An introduction to switched systems and their use in control is given, for instance, in [10]. In order to simplify the design process for the auxiliary system, the switching in the auxiliary system is limited to the switching among the state matrices $A_{\sigma,Aux} \in \{A_{1,Aux}, \ldots, A_{l,Aux}\}$, while the input matrix B_{Aux} and output matrix C_{Aux} are constant. The dynamics of the auxiliary system are given by

$$x_{Aux}(k+1) = A_{\sigma,Aux}x_{Aux}(k) + B_{Aux}u_{Aux}(k) \quad (3)$$
$$y_{Aux}(k) = C_{Aux}x_{Aux}(k) \quad (4)$$

where $x_{Aux} \in \mathbb{R}^{n_{Aux}}$, $u_{Aux} \in \mathbb{R}^{m_{Aux}}$ and $y_{Aux} \in \mathbb{R}^{p_{Aux}}$. The matrices $A_{\sigma,Aux}$, B_{Aux} and C_{Aux} have appropriate dimensions. $\sigma(k) \in \{1, \ldots, l\}$ denotes the arbitrary switching signal between the l subsystems.

The switched system (3)-(4) is stable for arbitrary switching [10], if there is a positive definite matrix P that satisfies

$$A_{\sigma,Aux}PA_{\sigma,Aux} - P \prec 0,\ \sigma = 1, \ldots, l \quad (5)$$

where $\prec$ denotes that the left side of (5) is negative definite.

In the next step the dynamics of the plant (1)-(2) and the auxiliary system (3)-(4) are coupled together to emulate the coupling found in most MIMO systems. Therefore the coupling matrices A_{Coup}, B_{Coup} and C_{Coup} are introduced. The extended system is then given by

$$\underbrace{\begin{bmatrix} x_{Sys}(k+1) \\ x_{Aux}(k+1) \end{bmatrix}}_{x(k+1)} = \underbrace{\begin{bmatrix} A_{Sys} & 0 \\ A_{Coup} & A_{\sigma,Aux} \end{bmatrix}}_{A_{\sigma,\ Comp}} \underbrace{\begin{bmatrix} x_{Sys}(k) \\ x_{Aux}(k) \end{bmatrix}}_{x(k)} + \underbrace{\begin{bmatrix} B_{Sys} & 0 \\ B_{Coup} & B_{Aux} \end{bmatrix}}_{B_{Comp}} \underbrace{\begin{bmatrix} u_{Sys}(k) \\ u_{Aux}(k) \end{bmatrix}}_{u(k)} \quad (6)$$

$$\underbrace{\begin{bmatrix} y_{Sys}(k) \\ y_{Aux}(k) \end{bmatrix}}_{y(k)} = \underbrace{\begin{bmatrix} C_{Sys} & 0 \\ C_{Coup} & C_{Aux} \end{bmatrix}}_{C_{Comp}} \begin{bmatrix} x_{Sys}(k) \\ x_{Aux}(k) \end{bmatrix} \quad (7)$$

where $A_{Coup} \in \mathbb{R}^{n_{Aux} \times n_{Sys}}$, $B_{Coup} \in \mathbb{R}^{n_{Aux} \times m_{Sys}}$ and $C_{Coup} \in \mathbb{R}^{p_{Aux} \times n_{Sys}}$. Since the auxiliary system has no access to the states of the original plant, the outputs of the plant are used as a substitute, since they are directly linked to the states of the original system through the output matrix, $y_{Sys} = C_{Sys}x_{Sys}$. To reduce the number of transmission channels needed to transmit y_{Sys} and u_{Sys} from the I/O to the auxiliary system the time-invariant reduction matrices $Q_{Red,u}$ and $Q_{Red,y}$ are introduced. The number of available communication channels for the transmission of y_{Sys} and u_{Sys} is denoted by θ_y and θ_u respectively. The dimensions of $Q_{Red,u}$ and $Q_{Red,y}$ are given by

$$\begin{aligned} &dim(Q_{Red,y}y_{Sys}) \overset{!}{=} \theta_y \Rightarrow dim(Q_{Red,y}) = \theta_y \times p_{Sys} \\ &dim(Q_{Red,u}u_{Sys}) \overset{!}{=} \theta_u \Rightarrow dim(Q_{Red,u}) = \theta_u \times m_{Sys} \\ &\quad with\ dim(X) = a \times b,\ X \in \mathbb{R}^{a \times b} \end{aligned}$$

To get the required size $Q_{Red,u}$ and $Q_{Red,y}$ have to be multiplied with matrices of appropriate dimensions. Therefore the time-invariant matrices $Q_{Exp1,y}$, $Q_{Exp2,y}$ and $Q_{Exp,u}$ are introduced and left multiplied with $Q_{Red,u}$ and $Q_{Red,y}$. This leads to the dimensions of $Q_{Exp1,y}$ being $n_{Aux} \times \theta_y$, of $Q_{Exp2,y}$ being $p_{Aux} \times \theta_y$ and of $Q_{Exp,u})$ being $n_{Aux} \times \theta_u$.

The introduction of the reduction and expansion matrices result in the coupling matrices

$$A_{Coup} = Q_{Exp1,y}Q_{Red,y}C_{Sys} \tag{8}$$

$$B_{Coup} = Q_{Exp,u}Q_{Red,u} \tag{9}$$

$$C_{Coup} = Q_{Exp2,y}Q_{Red,y}C_{Sys} \tag{10}$$

The reduction matrices are needed to use the information of all input and output signals of the original system for the coupling to strengthen the coupling and therefore increase the influence of the attacker on y_{Aux}. The number of I/O channels needed to transmit this information is also reduced, which saves money since additional I/O cards have to be bought. Therefore the input and output signals are multiplied by $Q_{Red,u}$ and $Q_{Red,y}$ respectively to reduce the number of required I/O channels. The expansion matrices are needed to get the right dimensions of the coupling matrices to describe the extended system as given in (6) and (7).

B. Design of the auxiliary system's matrices

As mentioned in section II, the auxiliary system should be stable and have similar properties as the plant itself. Therefore $A_{\sigma,Aux} \in \{A_{1,Aux}, \ldots, A_{l,Aux}\}$ should have similarly valued elements and poles as the matrix A_{Sys}. The matrices B_{Aux} and C_{Aux} should likewise be designed in such a way that they are similar to B_{Sys} and C_{Sys} respectively. The coupling matrices A_{Coup}, B_{Coup} and C_{Coup} should be designed so that the x_{Sys} and u_{Sys} have a similar influence on x_{Aux} and y_{Aux} as on x_{Sys} and y_{Sys}.

1) Truncated normal distribution: To generate the elements of the auxiliary system's matrices a truncated normal distribution (TND) [11, p.144] is used. This ensures that similar matrices are generated without values that significantly differ from the values they are based on, which may happen in a standard normal distribution. The matrices can also be generated by hand, but using a truncated normal distribution the process of obtaining the matrices can be automated.

The truncated normal distribution is given by

$$f(x;\mu,\sigma,a,b) = \frac{\frac{1}{\gamma}\phi(\frac{x-\mu}{\gamma})}{\Phi(\frac{b-\mu}{\gamma}) - \Phi(\frac{a-\mu}{\gamma})} \tag{11}$$

with a lower bound a, upper bound b ($b > a$), the location of the maximal probability μ and the squared scale γ^2. The probability density function of the standard normal distribution is $\phi(\xi) = \frac{1}{\sqrt{2\pi}}e^{(-\frac{1}{2}\xi^2)}$ and its cumulative distribution function is $\Phi(x) = \int_{-\infty}^{x}\phi(\xi)d\xi$. If $b = \infty$ then $\Phi(\frac{b-\mu}{\gamma}) = 1$ and if $a = -\infty$, then $\Phi(\frac{a-\mu}{\gamma}) = 0$.

2) Design of $A_{\sigma,Aux}$: For every subsystem of the auxiliary system a system matrix $A_{\sigma,Aux}$ has to be designed. Each matrix should be similar to A_{Sys}. The matrices $A_{\sigma,Aux}$ should therefore have eigenvalues in the same range as the ones in A_{Sys} and the elements should have similar values. Assume that the set of eigenvalues of A_{Sys} is $\Lambda = \{\lambda_1, \ldots, \lambda_{n_{Sys}}\}$.

To ensure a similar response to the original system the ratio of asymptotically decaying and decaying asymptotically decaying sinusoidal response components should be the same in the auxiliary system. The generation of the poles for $A_{\sigma,Aux}$ is split into two parts. The number of poles of $A_{\sigma,Aux}$ is equal to the number of states of the auxiliary system n_{Aux}. It is the sum of the number of real valued poles τ_{real} and twice the number of complex valued pole pairs τ_{comp}.

First the poles resulting in purely asymptotically decaying response components are obtained. The parameters of the TND to generate these poles are derived from the eigenvalues of A_{Sys} that are real and greater than 0, which result in asymptotically decaying response components. This set Λ_{real} is given by $\Lambda_{real} = \{\lambda_1, \ldots, \lambda_g\},\ \lambda_i \in \mathbb{R}_{\geq 0}$ and the parameters of the TND are given by

$$\begin{aligned} &a_{A,real} = min\{\Lambda_{real}\},\ b_{A,real} = max\{\Lambda_{real}\} \\ &\mu_{A,real} = \frac{1}{g}\sum_{i=0}^{g}\lambda_i,\ \lambda_i \in \Lambda_{real},\ \gamma_{A,real} = var\{\Lambda_{real}\} \end{aligned}$$

These are used to generate the set of real valued poles $\Delta_{real} = \{\delta_1, \ldots, \delta_{\tau_{real}}\}$.

The remaining poles are generated in two steps. In the first step the distance from the origin and in the second step the angle to the real axis are generated. For these poles, the poles of A_{Sys} that are not used in the previous step are considered. Since the complex poles only exist in complex conjugated pairs, only the poles with a positive imaginary part are used. This set is called Λ_{im} and given by

$$\Lambda_{im} = \{r_1e^{\mathrm{i}\varphi_1}, \ldots, r_he^{\mathrm{i}\varphi_h}\}; r_je^{\mathrm{i}\varphi_j} \in \Lambda \in \mathbb{C} \setminus \mathbb{R}_{>0},\ \varphi_j \leq \pi$$

with the set of angles $\Lambda_{ang} = \{\varphi_1, \ldots, \varphi_h\}$ and the set of radii $\Lambda_{rad} = \{r_1, \ldots, r_h\}$. With these two sets a set of complex poles for the system matrices similar to the poles of A_{Sys} can be generated and from this the set of complex conjugated poles.

The parameters used to get the angles are

$$\begin{aligned} &a_{A,ang} = min\{\Lambda_{ang}\},\ b_{A,ang} = max\{\Lambda_{ang}\} \\ &\mu_{A,ang} = \frac{1}{h}\sum_{i=0}^{h}\lambda_i,\ \lambda_i \in \Lambda_{ang},\ \gamma_{A,ang} = var\{\Lambda_{ang}\} \end{aligned}$$

The parameters used to get the radii are

$$\begin{aligned} &a_{A,rad} = min\{\Lambda_{rad}\},\ b_{A,rad} = max\{\Lambda_{rad}\} \\ &\mu_{A,rad} = \frac{1}{h}\sum_{i=0}^{h}\lambda_i,\ \lambda_i \in \Lambda_{radius},\ \gamma_{A,rad} = var\{\Lambda_{rad}\} \end{aligned}$$

With these variables the set of complex poles can be created

$$\Delta_{comp} = \{r_{im,1}e^{i\varphi_{im,1}}, r_{im,1}e^{-i\varphi_{im,1}}, \dots, r_{im,\tau_{comp}}e^{i\varphi_{im,\tau_{comp}}}, r_{im,\tau_{comp}}e^{-i\varphi_{im,\tau_{comp}}}\} = \{\delta_{\tau_{real}+1}, \dots, \delta_{\tau_{real}+2\tau_{comp}}\}$$

The set of eigenvalues for $A_{\sigma,Aux}$ is given by

$$\Delta = \Delta_{real} \cup \Delta_{complex} = \{\delta_1, \dots, \delta_{n_{Aux}}\} \quad (12)$$

Having generated the eigenvalues of the matrices $A_{\sigma,Aux}$, they can be used to generate a diagonal matrix $M_\sigma = diag(\delta_1, \delta_2, \dots, \delta_{n_{Aux}})$ with the desired eigenvalues. To fill the elements outside the main diagonal are non zero, but retain the original eigenvalues, the matrix can be multiplied with a unitary matrix on the right side and the inverse of the unitary matrix on the left side. The system matrix is then given by $A_{\sigma,Aux} = U^{-1}M_\sigma U$. The design of $A_{\sigma,Aux}$ has to be repeated until the desired number of state matrices for the switched system have been obtained.

The matrices obtained this way are asymptotically stable. This is a necessary and but not sufficient condition for the stability of the whole system for arbitrary switching [12]. For the whole system to be stable for arbitrary switching a common Lyapunov function (5) for all $A_{i,\,Aux}$, $i = 1, \dots, l$, has to be found. If there is no common Lyapunov function for all $A_{\sigma,Aux}$ other sets of $A_{\sigma,Aux}$ have to be created until a common Lyapunov function can be found.

3) Design of B_{Aux} and C_{Aux}: The matrices B_{Aux} and C_{Aux} are also designed using a truncated normal distribution. Contrary to the design of $A_{\sigma,Aux}$, the eigenvalues of theses matrices are not of interest for the stability of the system. Therefore B_{Aux} and C_{Aux} can be directly generated.

The distribution parameters for the generation of B_{Aux} are

$$a_B = min\{b_{i,j}\},\ b_B = max\{b_{i,j}\}$$
$$b_{i,j} \in B_{Sys},\ i \in \{1, \dots, n\},\ j \in \{1, \dots, m\}$$
$$\mu_B = \frac{1}{n\,m}\sum_{i=1}^{n}\sum_{j=1}^{m} b_{i,j},\ \sigma_B = var\{b_{i,j}\}$$

A truncated normal distribution with theses parameters is used to generate every element of $B_{Aux} \in \mathbb{R}^{n_{Aux} \times m_{Aux}}$.

To get the parameters for the distribution of C_{Aux}, the input matrix of the system C_{Sys} is used to create the parameters in the same way as B_{Sys} is used in the generation of B_{Aux}.

4) Design of the coupling matrices for the auxiliary system: For the design of A_{coup} the stability of the overall system matrix $A_{\sigma,Comp}$ has to be taken into account. An eigenvalue of $A_{\sigma,Comp}$ given in (6) satisfies the condition

$$\begin{aligned} &det(A_{\sigma,Comp} - \lambda I) \\ &= det(A_{Sys} - \lambda I) det(A_{\sigma,Aux} - \lambda I) = 0 \end{aligned}$$

thus the eigenvalues of A_{Coup} don't influence the stability of the overall system. Hence there are no restrictions on the eigenvalues of A_{Coup} and therefore not on $Q_{Exp1,y}$ and $Q_{Red,y}$. The matrices B_{Coup} and C_{Coup} don't influence the stability of the overall system either. Therefore the design of the reduction and expansion matrices $Q_{Red,y}$, $Q_{Red,u}$, $Q_{Exp1,y}$, $Q_{Exp2,y}$ and $Q_{Exp,u}$ are not restricted in any way other than their dimensions. The matrices can therefore be constructed using random variables similar to the construction of B_{Aux} and C_{Aux}.

The reduction and expansion matrices are generated from the same parameters used for the generation of B_{Aux} or C_{Aux}. The resulting absolute value of the elements of the coupling matrices (8)-(10) are larger than the desired absolute element values. This leads to the need for the matrices $Q_{Exp1,y}$, $Q_{Red,y}$, $Q_{Red,y}$, $Q_{Red,u}$ and $Q_{Exp1,y}$ to be scaled.

V. Attacker structure

In this section the attacker structure from [7] is extended to include the auxiliary system. If only the original system is attacked, the attack influences the output of the auxiliary system due to the coupling of these two systems. Therefore the auxiliary system has also to be taken into account by the attacker to stay undetected as long as possible.

The attacker calculates his output signal based on the extended system

$$\begin{aligned} \underbrace{\begin{bmatrix} x_{a,Sys}(k+1) \\ x_{a,Aux}(k+1) \end{bmatrix}}_{x_a(k+1)} &= \underbrace{\begin{bmatrix} A_{Sys} & 0 \\ A_{Coup} & A_{a,\,Aux} \end{bmatrix}}_{A_{a,Comp}} \underbrace{\begin{bmatrix} x_{a,Sys}(k) \\ x_{a,Aux}(k) \end{bmatrix}}_{x_a(k)} \\ &+ \underbrace{\begin{bmatrix} B_{Sys} & 0 \\ B_{Coup} & B_{Aux} \end{bmatrix}}_{B_{a,Comp}} \underbrace{\begin{bmatrix} u_{a,Sys}(k) \\ u_{a,Aux}(k) \end{bmatrix}}_{u_a(k)} \end{aligned} \quad (13)$$

$$\underbrace{\begin{bmatrix} y_{a,Sys}(k) \\ y_{a,Sys}(k) \end{bmatrix}}_{y_a(k)} = \underbrace{\begin{bmatrix} C_{Sys} & 0 \\ C_{Coup} & C_{Aux} \end{bmatrix}}_{C_{a,Comp}} \begin{bmatrix} x_{a,Sys}(k) \\ x_{a,Aux}(k) \end{bmatrix} \quad (14)$$

with the attacker's states x_a, outputs y_a and inputs u_a, and matrices $A_{a,Comp}$, $B_{a,Comp}$ and $C_{a,Comp}$. The matrices $A_{a,Aux}$ are the attacker's estimate of $A_{\sigma,Aux}$.

The dynamics of the extended system attacked by a covert attack are given by

$$x(k+1) = A_{\sigma,Comp}x(k) + B_{Comp}\left(u(k) + u_a(k)\right) \quad (15)$$
$$y^*(k) = C_{Comp}x(k) - y_a(k) \quad (16)$$

with the attacked input $u^*(k) = u(k) + u_a(k)$ and the attacked output $y^*(k)$.

VI. Design of the switched observer

To detect the covert attack a fault detection observer with a switched Luenberger observer as shown in [13] is used.

A. Construction of the switched observer

The design of the observer and the discussion of the error dynamics are taken from [13]. The residual is obtained by a full order observer using the system model given in (6)-(7). This results in the following observer

$$\begin{aligned} \hat{x}(k+1) &= A_{\sigma,Comp}\hat{x}(k) + B_{Comp}u(k) \\ &\quad + L_\sigma\left(y^*(k) - \hat{y}(k)\right) \end{aligned} \quad (17)$$
$$\hat{y}(k) = C_{Comp}\hat{x}(k) \quad (18)$$
$$r(k) = V(y^*(k) - \hat{y}(k)) \quad (19)$$

with the switched observer gain matrices L_σ, $\sigma = 1, \dots, l$ governed by the same switching signal $\sigma(k)$ as the system matrices $A_{\sigma,Aux}$ of the auxiliary system.

B. Residual in the attack free case

In the attack free case, where $u_a = 0$ and $y_a = 0$, let the state estimation error be

$$e(k) = \begin{bmatrix} e_{Sys}(k) \\ e_{Aux}(k) \end{bmatrix} = \begin{bmatrix} x_{Sys}(k) \\ x_{Aux}(k) \end{bmatrix} - \begin{bmatrix} \hat{x}_{Sys}(k) \\ \hat{x}_{Aux}(k) \end{bmatrix} \tag{20}$$

The residual dynamics are given by

$$e(k+1) = (A_{\sigma,Comp} - L_\sigma C_{Comp})\, e(k) \tag{21}$$

$$r(k) = V C_{Comp} e(k) \tag{22}$$

With the observer gain matrices L_σ designed to ensure the stability of $A_{\sigma,Comp} - L_\sigma C_{Comp}$, the residual converges to zero in the attack free case.

C. Residual dynamics in case of a covert attack

With the system model (1)-(2), the attacker's system model (13)-(14) and the switched observer (17)-(19) the residual in case of an attack can be calculated. The attacker is assumed to have obtained the system matrix of the active subsystem of the auxiliary system and all constant matrices.

The state estimation error of the attacked extended system (15)-(16) and its dynamics are given by

$$e(k) = x(k) - x_a(k) - \hat{x}(k) \tag{23}$$

$$\begin{aligned} e(k+1) &= A_{\sigma,Comp} x(k) - A_{a,Comp} x_a(k) - A_{\sigma,Comp} \hat{x}(k) \\ &\quad - L_\sigma C_{Comp} \left(x(k) - x_a(k) - \hat{x}(k) \right) \end{aligned} \tag{24}$$

1) Attacker with perfect system model and σ: If the attacker has the complete system model (i.e. $A_{a,Aux} = A_{\sigma,Aux}$) and knows $\sigma(k)$, the state estimation error in case of a covert attack given in (24) becomes $e(k+1) = (A_{\sigma,Comp} - L_\sigma C_{Comp})\, e(k)$, where $e(k+1)$ converges to zero if $A_{\sigma,Comp} - L_\sigma C_{Comp}$ is stable.

The residual then becomes $r(k) = V C_{Comp} e(k)$ and converges to zero if the state estimation error converges to zero. This shows that an attacker with a perfect system model and knowledge of $\sigma(k)$ can not be detected.

2) Attacker with imperfect system model: If the attacker's system matrix for the auxiliary system is not $A_{\sigma,Aux}$ (i.e. $A_{a,Aux} \neq A_{\sigma,Aux}$), the state estimation error (24) is indeed

$$\begin{aligned} \begin{bmatrix} e_{Sys}(k+1) \\ e_{Aux}(k+1) \end{bmatrix} &= (A_{\sigma,Comp} - L_\sigma C_{Comp}) \begin{bmatrix} e_{Sys}(k) \\ e_{Aux}(k) \end{bmatrix} \\ &\quad + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & A_{\sigma,Aux} - A_{a,Aux} \end{bmatrix}}_{\Delta A_{\sigma,Comp}} \begin{bmatrix} x_{a,Sys}(k) \\ x_{a,Aux}(k) \end{bmatrix} \end{aligned} \tag{25}$$

$\Delta A_{\sigma,Comp}$ is the attacker's model error of the state matrix.

In this case the residual dynamics become

$$\begin{aligned} \begin{bmatrix} r_{Sys}(k+1) \\ r_{Aux}(k+1) \end{bmatrix} &= V C_{Comp} \left(A_{\sigma,Comp} - L_\sigma C_{Comp} \right) \\ &\quad \cdot \begin{bmatrix} e_{Sys}(k) \\ e_{Aux}(k) \end{bmatrix} + V C_{Comp} \Delta A_{\sigma,Comp} \begin{bmatrix} x_{a,Sys}(k) \\ x_{a,Aux}(k) \end{bmatrix} \end{aligned} \tag{26}$$

As can be clearly seen in (26), the residual r_{Aux} of the auxiliary system will not converge to zero due to the error $\Delta A_{\sigma,Comp}$ in the system model of the attacker and can therefore be detected.

D. Selection of the observer gain matrices L_σ

Assume that all subsystems $(A_{\sigma,Comp}, C_{Comp})$ with $\sigma(k) \in \{1, \ldots, l\}$ are observable. The observer estimation error then converges asymptotically to zero [13], if there exists a symmetric positive definite matrix P that satisfies

$$\begin{aligned} &(A_{\sigma,\,Comp} - L_\sigma C_{Comp})^T P (A_{\sigma,\,Comp} - L_\sigma C_{Comp}) \\ &- P \prec 0,\ \sigma = 1, \ldots, l \end{aligned} \tag{27}$$

The assumption that all subsystems $(A_{\sigma,\,Comp},\ C_{Comp})$ with $\sigma = 1, 2, \ldots, l$ are observable is necessary to guarantee that each inequality in (27) may admit a solution for P.

Since P and L_σ influence each other, they have to be found together. With the substitution $R_\sigma = P L_\sigma$ the inequalities of (27) are transformed using the Schur complement to

$$\begin{aligned} &\begin{pmatrix} P & P A_{\sigma,Comp} - R_\sigma C_{Comp} \\ (P A_{\sigma,Comp} - R_\sigma C_{Comp})^T & P \end{pmatrix} \\ &\succ 0,\ \sigma = 1, 2, \ldots, l \end{aligned} \tag{28}$$

The inequalities of (28) fit the so-called linear matrix inequalities (LMI) framework and can be solved by means of convex programming algorithms. An introduction to LMIs for control purposes can be found in [14].

After finding P and R_σ that satisfy the inequalities of (28), the observer gain matrices L_σ of the switched observer can be determined by $L_\sigma = P^{-1} R_\sigma,\ \sigma = 1, \ldots, l$.

E. Residual evaluation

The decision logic for the attack detection is

$$\begin{cases} \|r_{Aux}\| > J_{th} \Rightarrow & System\ is\ under\ attack \\ \|r_{Aux}\| \leq J_{th} \Rightarrow & System\ is\ attack\ free \end{cases}$$

where the threshold J_{th} can be determined as the maximum fluctuation in the attack free case, i.e.

$$J_{th} = \max_{u_a = 0, y_a = 0} \|r_{Aux}\| \tag{29}$$

F. Remarks for the real-world implementation

The switching of the Luenberger observer and the auxiliary system have to be synchronized. For this switching function a pseudo random number generator with the same seed [15] in the observer and the auxiliary system with a time synchronization using radio transmitted time signals like the Global Positioning System [16] can be used.

VII. Simulation Example

The proposed method to detect covert attacks is shown with the example of the quadruple tank process [17]. The non-minimum phase system at working point $w = [12.6\ 13.0]$ is discretized using zero-order hold with $T_s = 1s$ to

$$A = \begin{bmatrix} 0.9843 & 0 & 0.0251 & 0 \\ 0 & 0.9892 & 0 & 0.0175 \\ 0 & 0 & 0.9747 & 0 \\ 0 & 0 & 0 & 0.9823 \end{bmatrix},$$

$$B = \begin{bmatrix} 0.0478 & 0.0010 \\ 0.0005 & 0.0348 \\ 0 & 0.0765 \\ 0.0554 & 0 \end{bmatrix},\ C = \begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \end{bmatrix},$$

Using the methods given in Section IV-B the system matrices of the auxiliary system are obtained with $l = 3$, $n_{Aux} = 2$, $m_{Aux} = 1$ and $p_{Aux} = 2$ and result in

$$A_{Aux,1} = \begin{bmatrix} 0.98267 & 3.06 \cdot 10^{-5} \\ 3.06 \cdot 10^{-5} & 0.98262 \end{bmatrix},$$

$$A_{Aux,2} = \begin{bmatrix} 0.98273 & 1.93 \cdot 10^{-5} \\ 1.93 \cdot 10^{-5} & 0.98263 \end{bmatrix},$$

$$A_{Aux,3} = \begin{bmatrix} 0.98266 & -9.49 \cdot 10^{-6} \\ -9.49 \cdot 10^{-6} & 0.98265 \end{bmatrix},$$

$$B_{Aux} = \begin{bmatrix} 0.0282 \\ 0.0271 \end{bmatrix}, \; C_{Aux} = \begin{bmatrix} 0.0615 & 0.1036 \\ 0.1756 & 0.0970 \end{bmatrix}$$

The matrices $A_{Aux,\sigma}$ do not differ much from one another since the eigenvalues of A are in very close vicinity.

The reduction and expansion matrices with $\theta_u = \theta_y = 1$ are given by

$$\begin{matrix} Q_{Red,y} = \begin{bmatrix} 0.0407 & 0.1487 \end{bmatrix}, \\ Q_{Red,u} = \begin{bmatrix} 0.1609 & 0.0150 \end{bmatrix}, \end{matrix} Q_{Exp1,y} = \begin{bmatrix} 0.1582 \\ 0.1547 \end{bmatrix},$$

$$Q_{Exp2,y} = \begin{bmatrix} 0.1670 \\ 0.0961 \end{bmatrix}, Q_{Exp,u} = \begin{bmatrix} 0.1396 \\ 0.1291 \end{bmatrix}$$

The observer gain matrices are calculated using (28).

The inputs of the original system are a step signal and a sinusoidal signal and for the auxiliary system a step signal. The attacker input signal u_a starts at $k = 1000s$ and is comprised of ramp signals with different slopes bounded by 0.5. x_{Sys} and x_{Aux} are initialized to 0.

The attacker is assumed to have perfect system knowledge at $k = 0s$. Therefore $A_{a,Comp} = A_{1,Comp}$, $B_{a,Comp} = B_{Comp}$ and $C_{a,Comp} = C_{Comp}$.

In the attack free case, the residual is close to zero in the simulation. It is not influenced by the switching between the subsystems of the auxiliary system at $k = 3000s$ and $k = 7000s$. In case of a covert attack, the residual r of the extended system is depicted in Figure 2. The residual of the original system, r_{Sys} (orange and blue lines, barely visible behind the orange line), is close to zero in case of a covert attack. The residual of the auxiliary system, r_{Aux} (purple and yellow lines) are close to zero when the attack starts at $k = 1000s$. At the time instant $k = 3000s$ the second subsystem is activated and the attack can be seen in the residual.

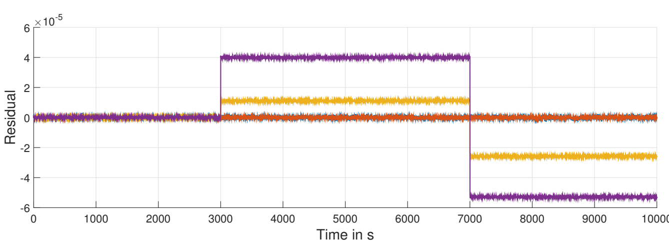

Fig. 2. Residual in case of a covert attack

VIII. Conclusion

It has been shown that the presented approach can be used successfully to detect covert attacks. By attaching the auxiliary system via analogue connections to the I/O of the system it can not be attacked through the network. Due to the coupling with the original system the attacker can not ignore the auxiliary system. The effective functioning in the presence of noise and disturbances can be shown.

An attacker with an initially perfect system model can only be detected after another subsystem is activated in the auxiliary system. The detection speed is therefore dependent on the speed in which the subsystems switch. The plant can still be controlled by the original controller. The auxiliary system has to be controlled by a controller that can handle the switched system dynamics.

The focus of this paper is on the detection of covert attacks, but the proposed approach can also be used to detect other attacks like replay attacks.

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